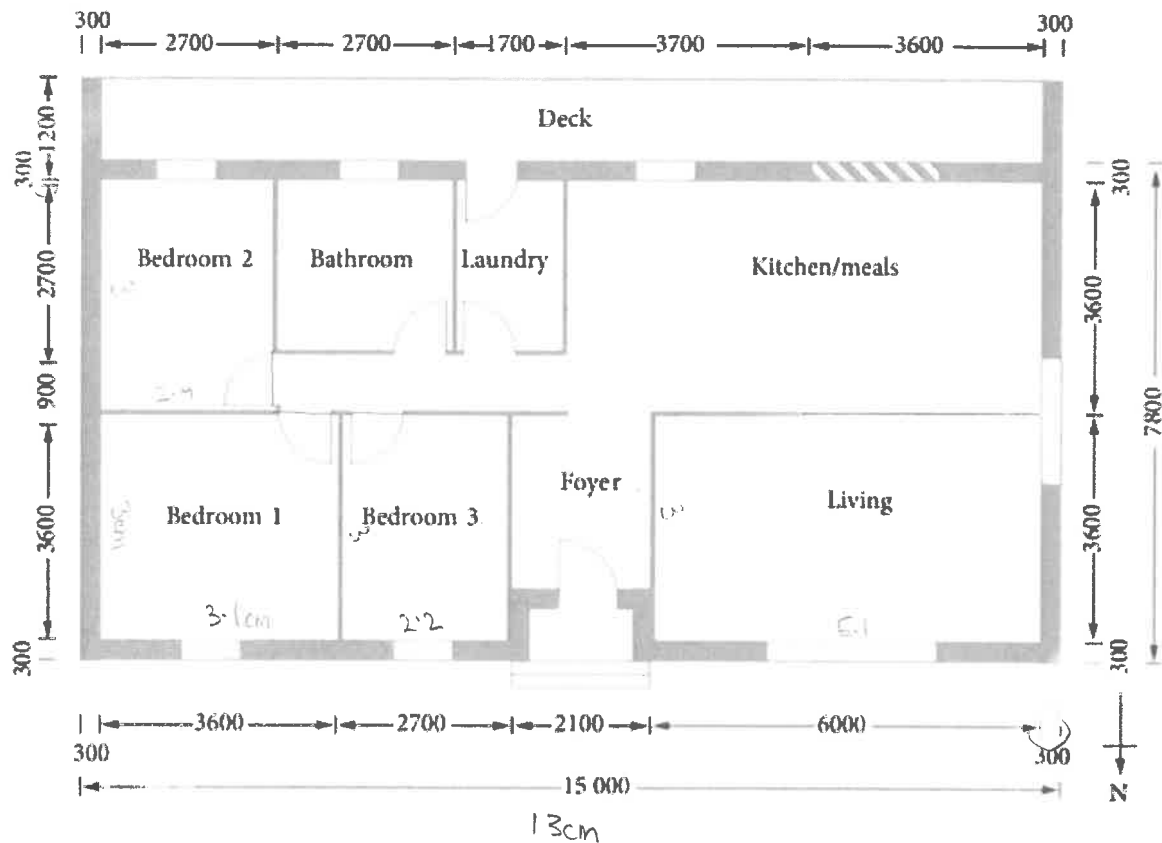


Week 7 Mathematical Thinking Process task

Fran and Mike are building a house according to the plan below. They need to order carpet for the living room and three bedrooms, allowing for an extra 10% for wastage. How many square meters of carpet will they need? Answer correct to the nearest square metre.



$$1\text{cm} = 1.15\text{m}$$

bedroom 1

$$= 3.1 \times 3$$

$$= 3.596 \times 3.48$$

$$= 12.51\text{m}^2$$

bedroom 2

$$= 3 \times 2.4$$

$$= 3.48 \times 2.784$$

$$= 9.69\text{m}^2$$

bedroom 3

$$= 3 \times 2.2$$

$$= 3.48 \times 2.552$$

$$= 8.88\text{m}^2$$

living room

$$= 3 \times 5.1$$

$$= 3.48 \times 5.916$$

$$= 20.59\text{m}^2$$

$$12.51 + 9.69 + 8.88 + 20.59$$

$$= 51.67\text{m}^2 \text{ total}$$

+ 10% extra

$$\text{nearest } \text{m}^2 \\ = 57\text{m}^2$$

$$51.67 \times 1.1 \\ = 56.83$$

Mathematical Thinking Process Planning Sheet

Interpret the task	to determine the amount of carpet needed for bedrooms 1, 2 and 3
Identify the mathematics involved	Scale drawings percentages = area connecting measurements
Analyse information/data (you may need to use other sources)	no extra information needed all assistance given when
Apply existing mathematical knowledge and strategies	refer to task page to view step by step working out.
Verify the reasonableness of your answer	faced difficulties as the scale was not given and the measurements are not all equal therefore the scale used is only an estimation $57m^2$ is reasonable due to the size of the house and the mid level number.
Communicate findings	See back page

Answer/Findings/ Solution:

The purpose of this investigation is to determine the number of square metres of carpet required to carpet three bedrooms and the living room. To achieve this number, the use of